

# SYSC 4504: Fundamentals of Web Development

## Programming Assignment 3: PHP, SQL

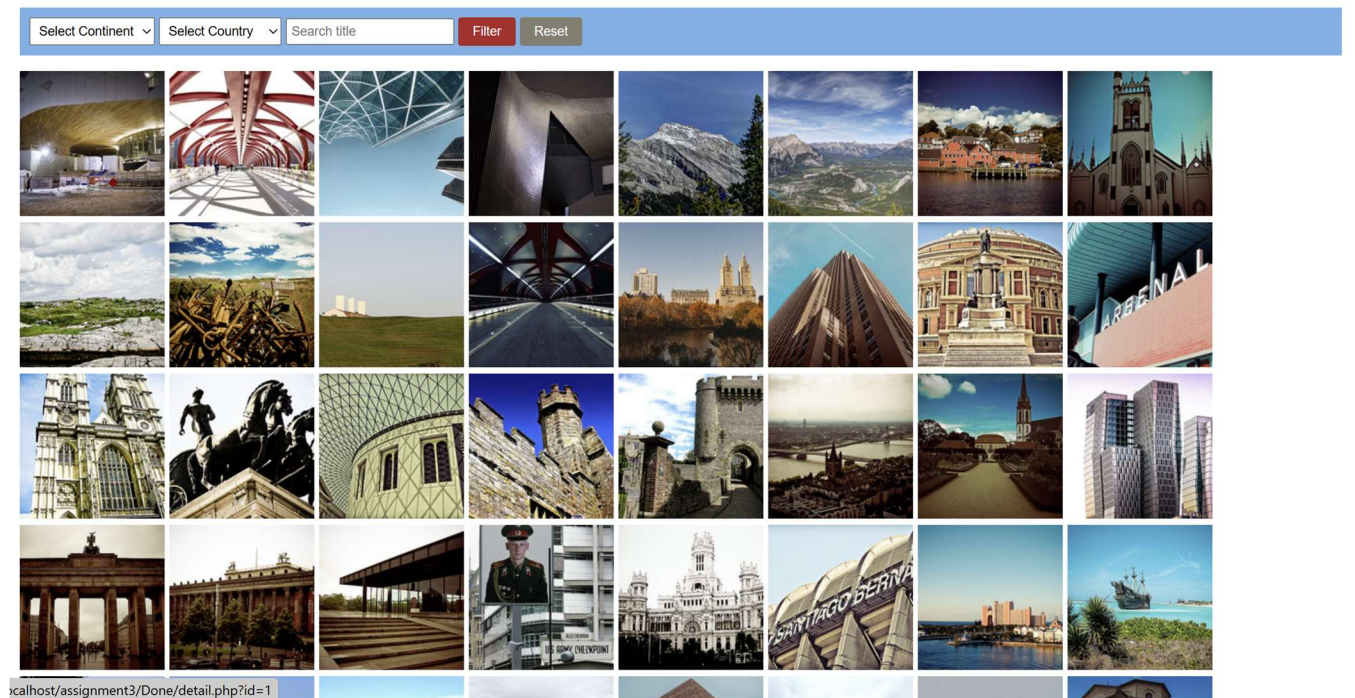
Due: March 27, 1 PM

### Assignment Description

This assignment will introduce server-side programming and is more complex than the first two ones. So you should budget more time for it. To get started, you need to have both the Apache webserver and the MySQL database from the XAMPP up and running, as you will be writing PHP code. You also need to load the travel database into MariaDB, available as part of the initial set of files, named `travel-3rd.sql`. Finally, create a user account (with at least read privileges but probably all data privileges), using the following credentials:

- Name: `testuser`
- Password: `mypassword`

Brightspace has an initial set of files, including two PHP files `assignment3.php` and `detail.php`. The first PHP resource, once completed, will show a collection of thumbnails, similar to the image below. It also contains a filter area on the top to reduce the number of thumbnails displayed on the page.



Once you click on a thumbnail image, `detail.php` will be invoked, which displays additional information about an image. For example, clicking on the first image above would yield the following display:



## Calgary Central Library

Calgary, Canada

The Calgary Central Library, also known as the Calgary New Central Library (NCL), is a public library in Calgary, Alberta, Canada, and the flagship branch of the Calgary Public Library. It is located in the Downtown East Village neighborhood and opened on November 1, 2018, replacing the existing central branch in Downtown Calgary. The four-storey building cost \$245 million to construct and was designed by American-Norwegian architecture firm Snøhetta and Canadian firm DIALOG after the two firms' joint bid won a design competition in 2013. Their design features an oval-like form and an interior with a large central atrium with a skylight. The building is elevated one floor above street level to accommodate a light rail trackway below as well as a public plaza.

### Creator

[Robert Boschman](#)

### Camera

Nikon D810  
ISO 450 | f1.8 | 1/40 sec

### Colors



Complete the following tasks with the files you were given as a starting point:

1. You will need to retrieve information from three tables: `continents`, `countries`, and `imagedetails`.
2. Display every image in the `imagedetails` table (i.e., as the `<main>` body of `assignment3.php`). The `Path` field contains the filename of the image. Each image should be a link to `detail.php` with the `ImageID` field passed as a query string. The supplied `detail.php` page contains sample markup for a single photo. You will need to construct a SQL query that joins data from the `imagedetails` table, the `countries` table, and the `cities` table. The camera and color information shown in the above screenshot are from the `Exif` and `Colors` fields and contain `json` data. You can use the `json_decode()` function to convert this `json` data into a PHP object.
3. The filter section near the top of the page will be used to filter/reduce the number of images displayed in the image list. The user will be able to display only those images from a specific `continent`, `country`, or `images` whose `Title` field contains a search word after the user clicks the `Filter` button.
4. You will need to display every record from the `continents` tables within the `<select>` list that appears in the filter section near the top of the page. Each `<option>` element should display the `ContinentName` field; the `ContinentCode` field should be used for the option value.
5. For the `Countries` `<select>` list, you will display only those countries that have a matching record in the `imagedetails` table. This will require an `INNER JOIN` along with a `GROUP BY`.
6. When the user clicks the `Filter` button, the page should display only those images whose `CountryCodeISO` or `ContinentCode` or `Title` fields match the specified valued in the filter area. For the `Title` field, match any records whose `Title` field contains whatever was entered into the search box (hint: use `SQL like` along with the wildcards character). When a user selects multiple filter options, filter only on the first one (i.e., `Continent` before `Countries` before `Title`), no need to combine the filter options.

## Guidance and Testing

1. Break this down into smaller steps. A good starting point would be to get your PHP page to read and display data from the `continents` table. Then do the same for the `countries` and `imagedetails` tables.

2. The styling has been already provided for you. Examine the sample markup within the supplied `<template>` elements.
3. Get the list of images to display correctly, then implement the details page.
4. Finally, add in the filter functionality.

Note: similar to Assignment 2, we are accessing data stored on the textbook authors' website from a page you downloaded from your local computer. To prevent issues related to CORS or Cross-Origin Resource Sharing, install the appropriate "Allow CORS" extension for the browser of your choice.

## Submission Requirements

Submit your assignment using Brightspace, under Assignment 3 Submission for the course. Create a single ZIP archive of all files that make up your solution, including the files we provided, and name that archive `Assignment3.zip`. Once the TAs unzip your submission, they should simply be able to access `assignment3.php` with a browser of their choice to test your submission. DO NOT submit your solution as a RAR or any other form of archive, nor submit individual files.

Marks will be based on:

- Completeness of your submission
- Correct solution to the problem
- Adhering to the submission requirements

The due date is based on the time of the Brightspace server and will be strictly enforced. If you are concerned about missing the deadline: multiple submissions are allowed. So you can always submit a (partial) solution early, and resubmit an improved solution later. This way, you will reduce the risk of running late, for whatever reason (slow computers/networks, unsynchronized clocks, failure of the Internet connection at home, etc.).